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Unit – 1 : Session – 12 : SLO – 1 | System Programs vs System Calls — Diagram Keywords

■ I. Diagram Keywords (Layered View)

- Top layer: USER PROGRAMS / APPLICATIONS
- Next: SYSTEM PROGRAMS (utilities: shells, editors, compilers, loaders)
- Next: SYSTEM CALL INTERFACE (API: open, read, write, fork, exec)
- Core: KERNEL (process management, memory management, file system, device drivers)
- Bottom: HARDWARE (CPU, memory, devices)

■ II. Distinction

- SYSTEM CALLS: the programmatic interface between a process and the kernel (e.g. fork, read).
- SYSTEM PROGRAMS (utilities): executable programs shipped with the OS that USE system calls to
- provide user-facing functionality (e.g. ls, cp, gcc, top). They run in user mode, not kernel mode.